SFI Challenge
Solution Proposal

TEAM Karnex/Thunder Brain

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GitHub repository link: https://github.com/Zimgo012/SFI-Challenge

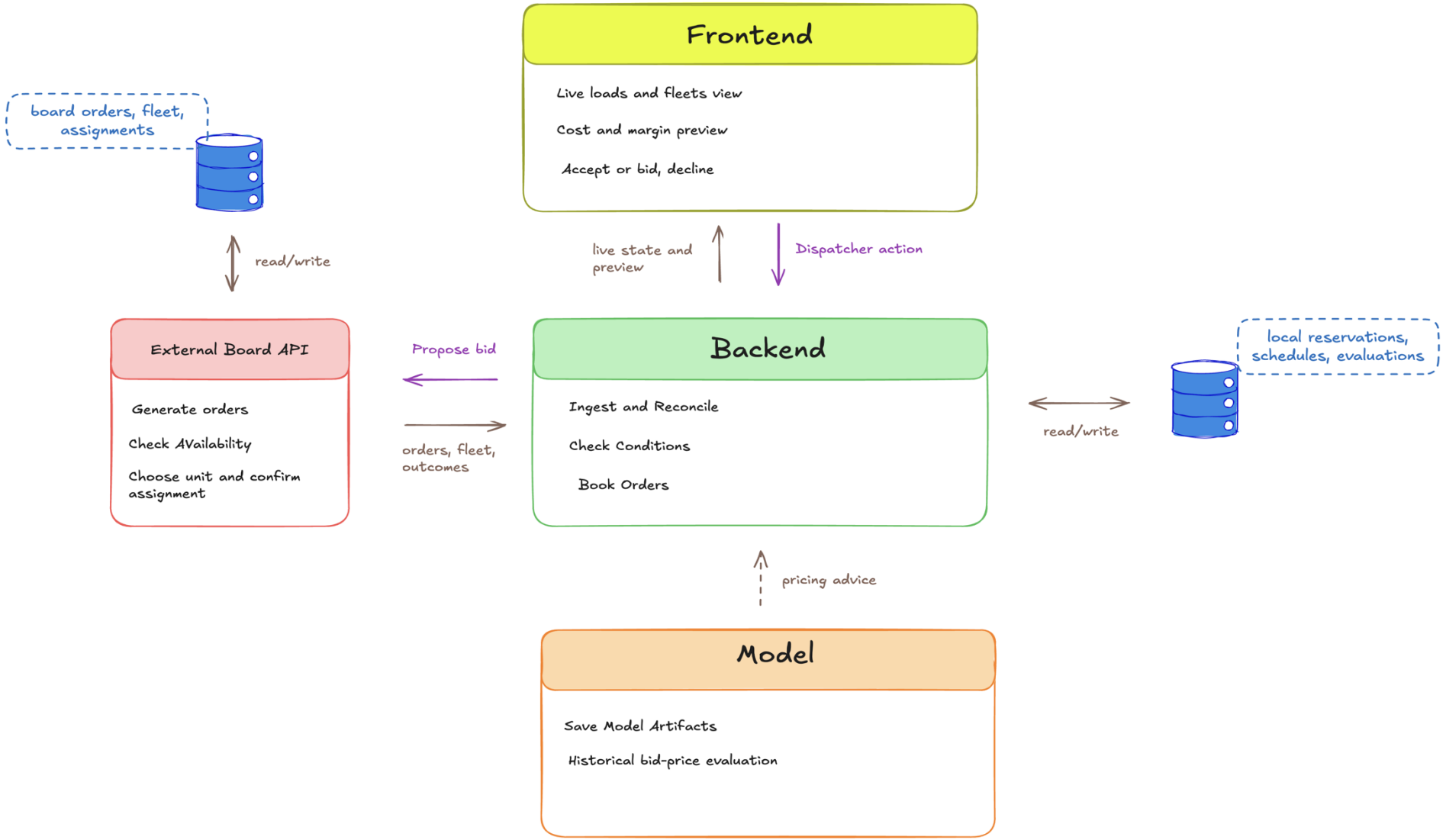
The Business Problem

- Trucks driving empty waste fuels, time, and money
- Conflicting schedules cause double-booking and missed pickups.
- Winning shipments does not guarantee profitable deliveries.
- Expiring offers demand fast, informed decisions.

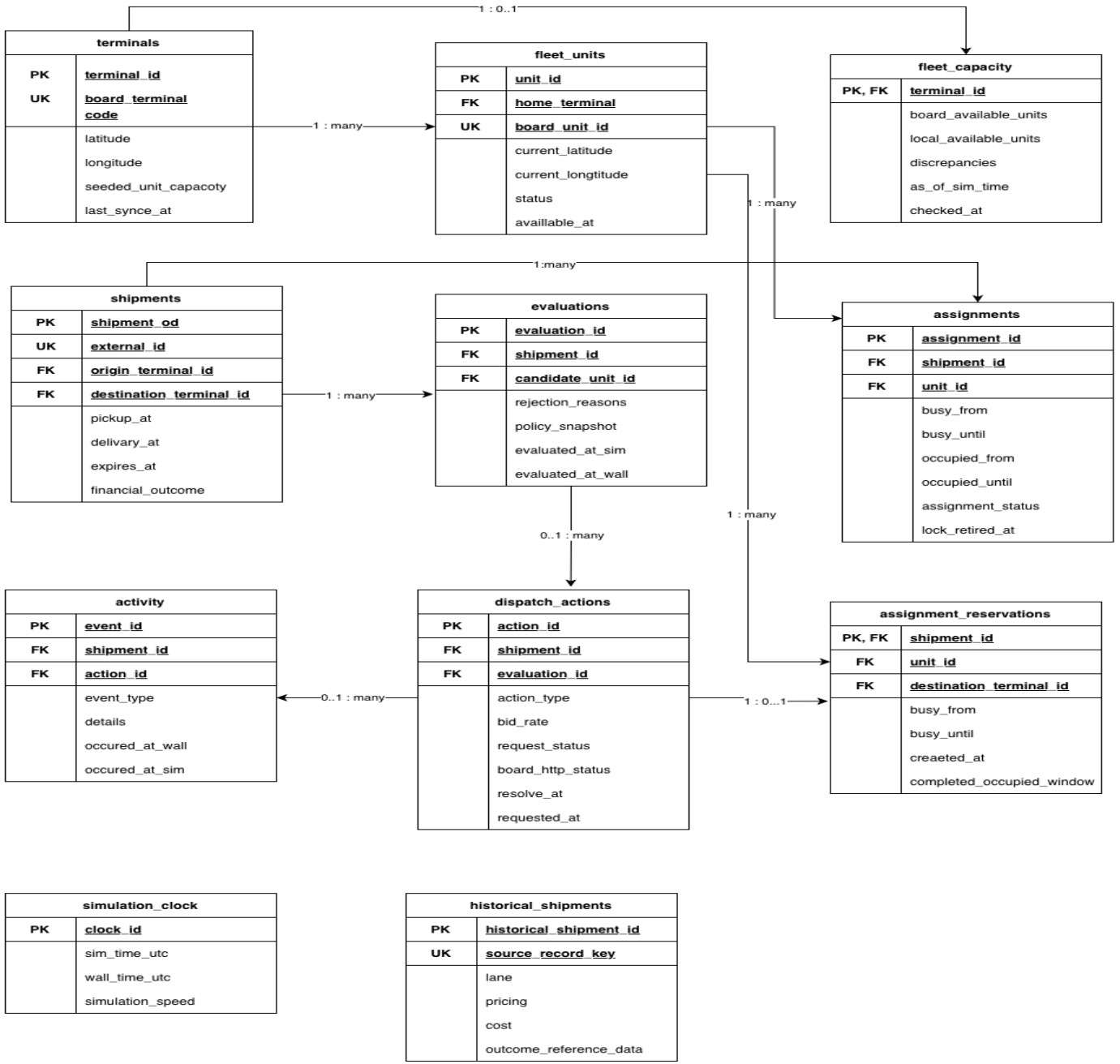
The Challenge Goal

- Reduce empty miles by chaining compatible loads.
- Match shipments with available trucks and conflict-free schedules.
- Protect profit margins before accepting or bidding.
- Use live data and guarded automation to capture worthwhile opportunities.

High-level System Architecture



ERD


1 : 0..1
terminals
PK terminal_id
UK board_terminal code
latitude
longitude
seeded_unit_capacoty
last_synce_at
1 : many
fleet_units
PK unit_id
FK home_terminal
UK board_unit_id
current_latitude
current_longtitude
status
availlable_at
1 : many
fleet_capacity
PK, FK terminal_id
board_available_units
local_available_units
discrepancies
as_of_sim_time
checked_at
1:many
shipments
PK shipment_od
UK external_id
FK origin_terminal_id
FK destination_terminal_id
pickup_at
delivary_at
expires_at
financial_outcome
1 : many
evaluations
PK evaluation_id
FK shipment_id
FK candidate_unit_id
rejection_reasons
policy_snapshot
evaluated_at_sim
evaluated_at_wall
1 : many
assignments
PK assignment_id
FK shipment_id
FK unit_id
busy_from
busy_until
occupied_from
occupied_until
assignment_status
lock_retired_at
0..1 : many
activity
PK event_id
FK shipment_id
FK action_id
event_type
details
occured_at_wall
occured_at_sim
0..1 : many
dispatch_actions
PK action_id
FK shipment_id
FK evaluation_id
action_type
bid_rate
request_status
board_http_status
resolve_at
requested_at
1 : 0...1
assignment_reservations
PK, FK shipment_id
FK unit_id
FK destination_terminal_id
busy_from
busy_until
creaeted_at
completed_occupied_window
simulation_clock
PK clock_id
sim_time_utc
wall_time_utc
simulation_speed
historical_shipments
PK historical_shipment_id
UK source_record_key
lane
pricing
cost
outcome_reference_data

What the Backend Delivers

1. Synchronize orders, fleet, terminals, and simulated time through WebSockets and REST reconciliation.

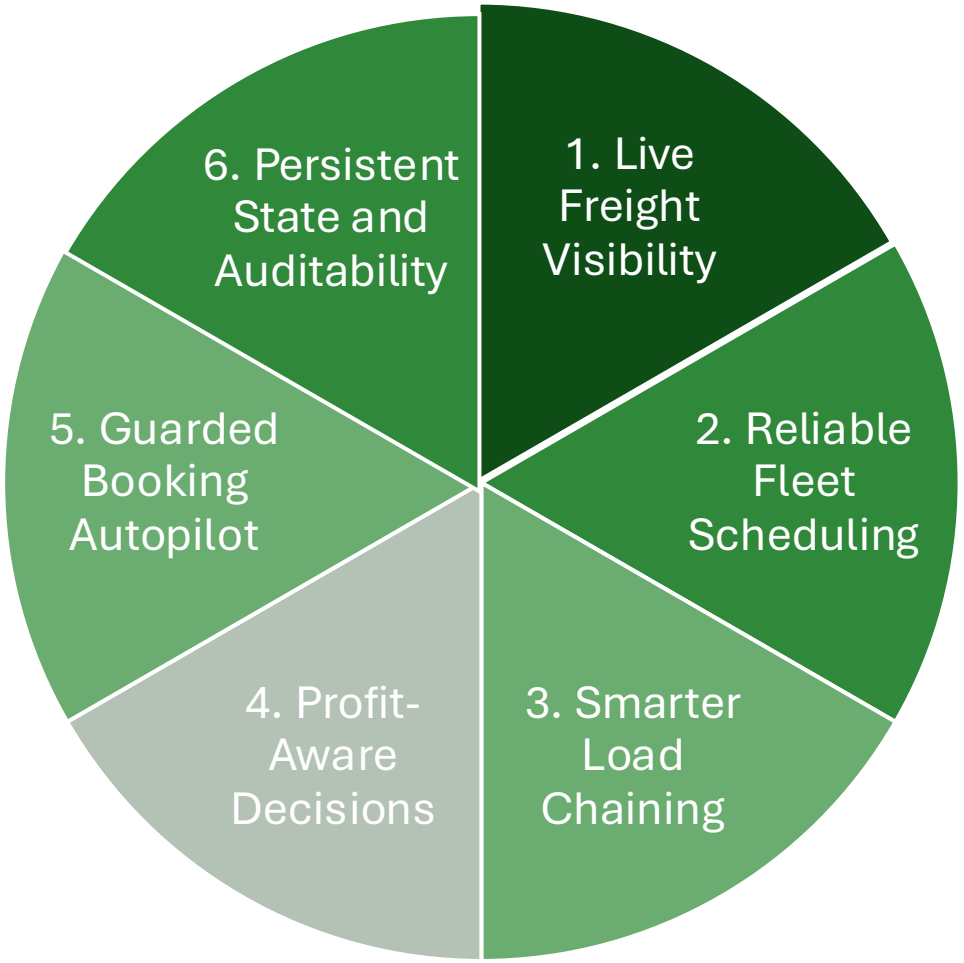
2. Check truck availability and full occupied windows, then reserve capacity atomically to prevent double-booking.

3. Project each truck's position after delivery and calculate its empty travel to the next pickup.

4. Calculate loaded and deadhead costs, expected profit, margin, and minimum profitable bids before booking.

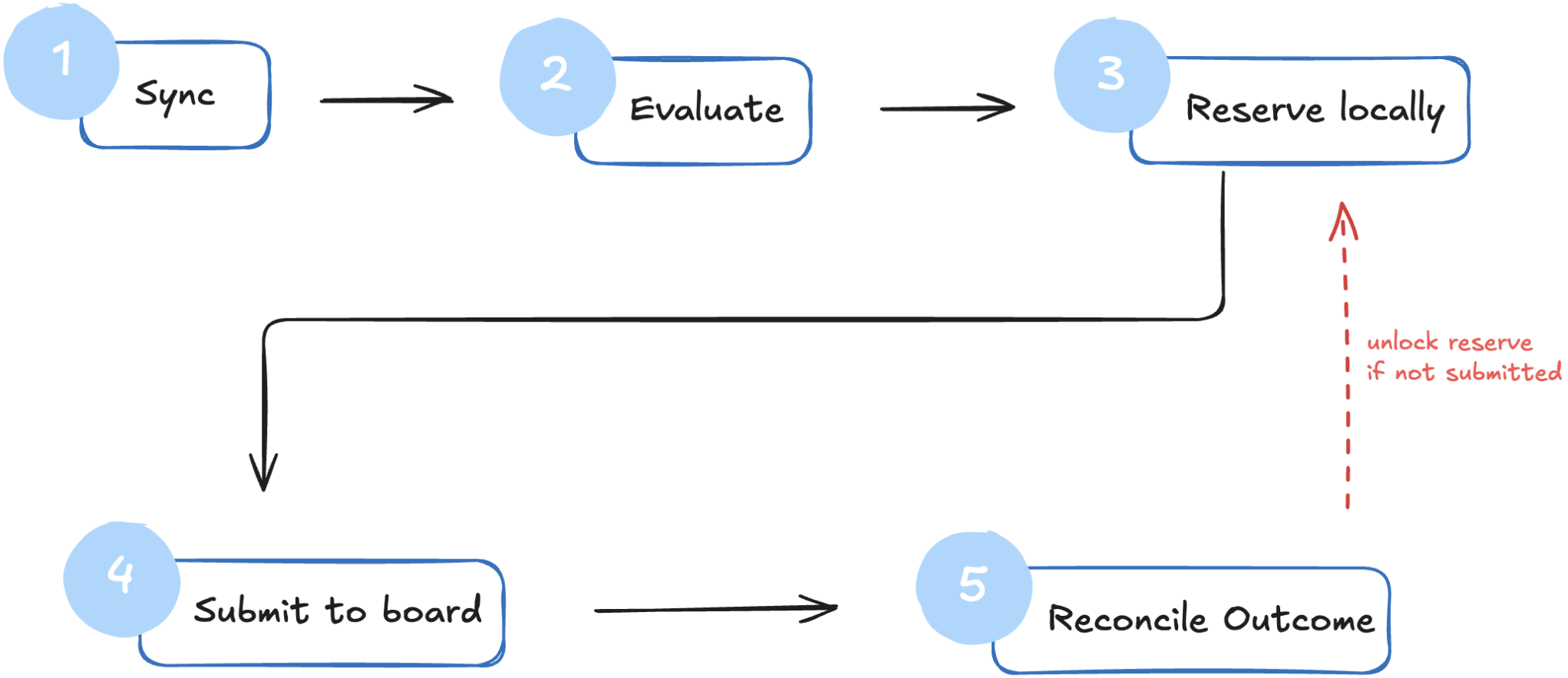
5. Support manual actions and model-selected automatic bookings, with fresh checks and configurable limits.

6. Store assignments, evaluations, actions, and outcomes in our own database; reconcile missed events and preserve decision history.



Backend workflow

High-level Backend Workflow



Capture Live Opportunities

- Sync orders, fleet, terminals, and simulated time.

Check Operational Feasibility

- Verify truck availability, schedules, weight, location, and expiry.

Calculate Expected Profit

- Estimate selected-truck deadhead, total cost, margin, and bid floor.

Choose the Booking

- Support a dispatcher’s decision or an authorized autopilot selection.

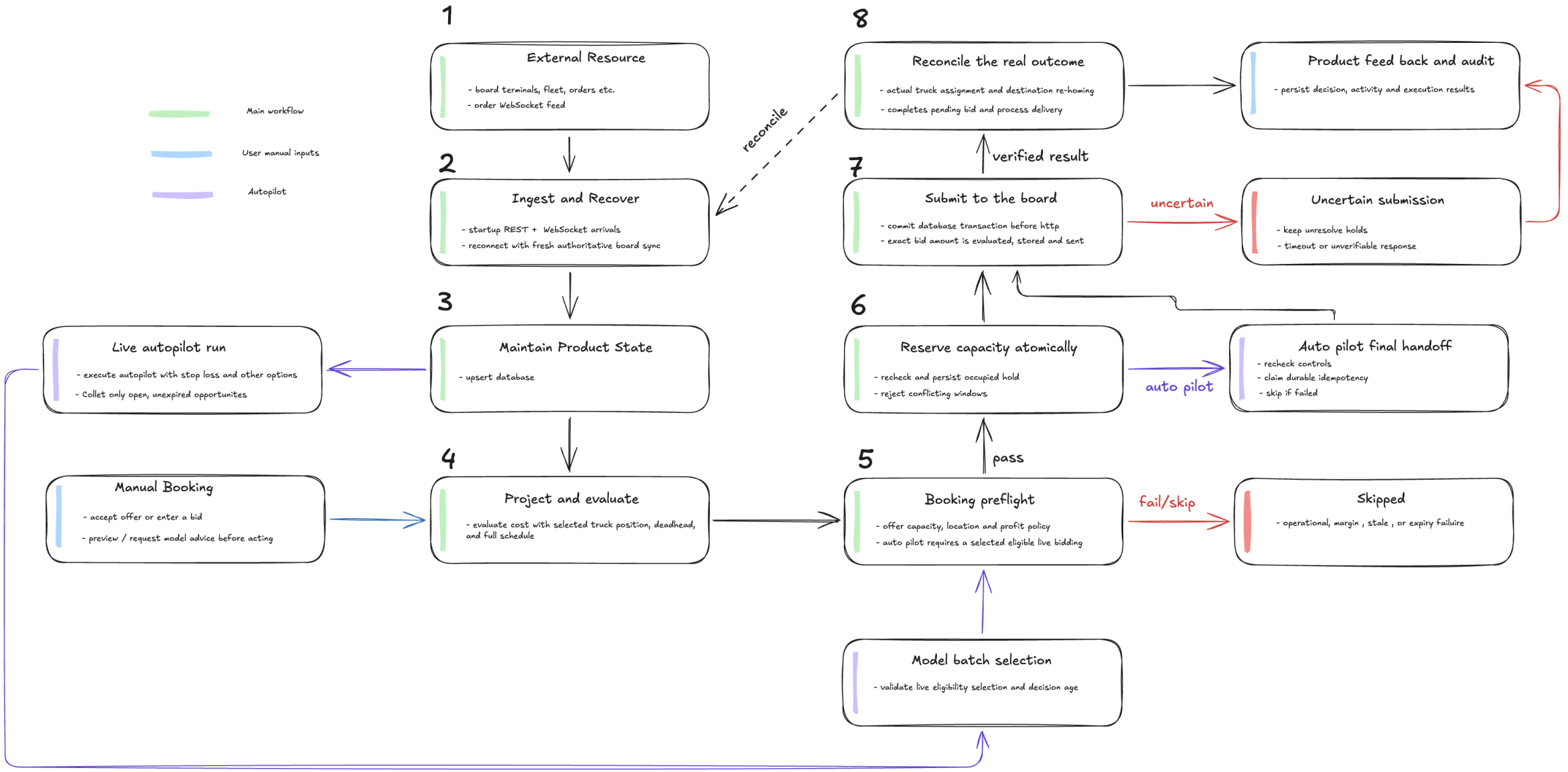
Reserve and Submit

- Reserve capacity atomically, recheck applicable conditions, and call the board.

Track and Reconcile

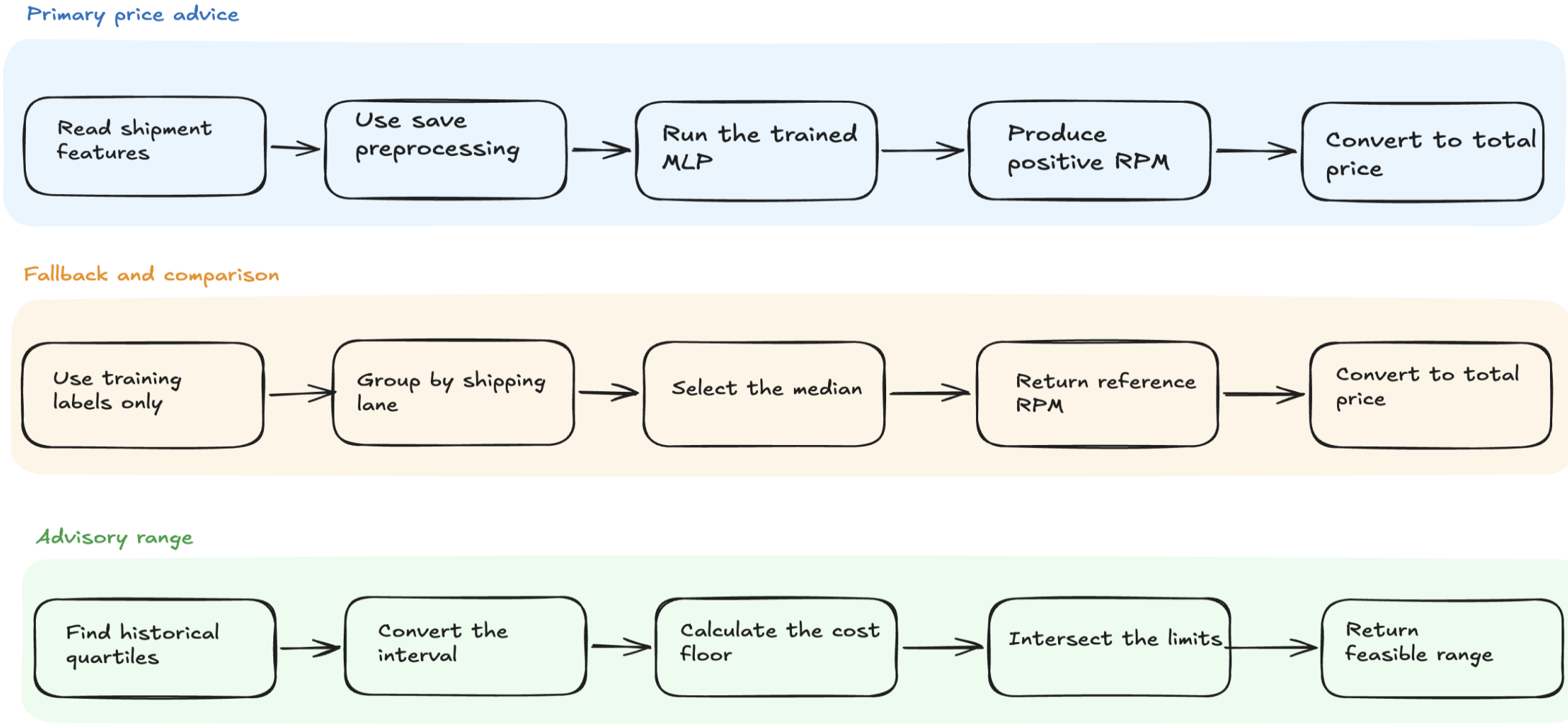
- Confirm awards, assignments, transit, and completion; update capacity and audit history.

Backend detailed workflow



Price Analyzer

Model Analyzer Step

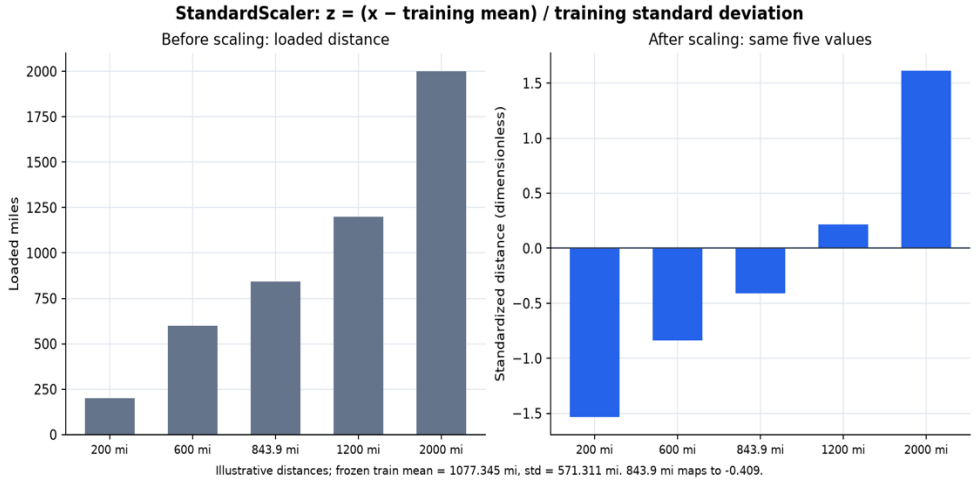


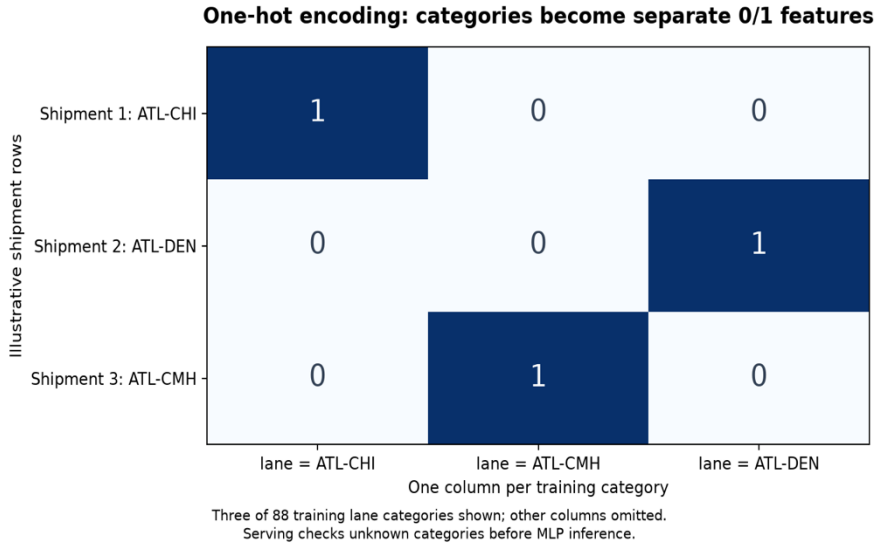
How the model train

Preparing shipment information

StandardScaler puts numeric values onto comparable scales.

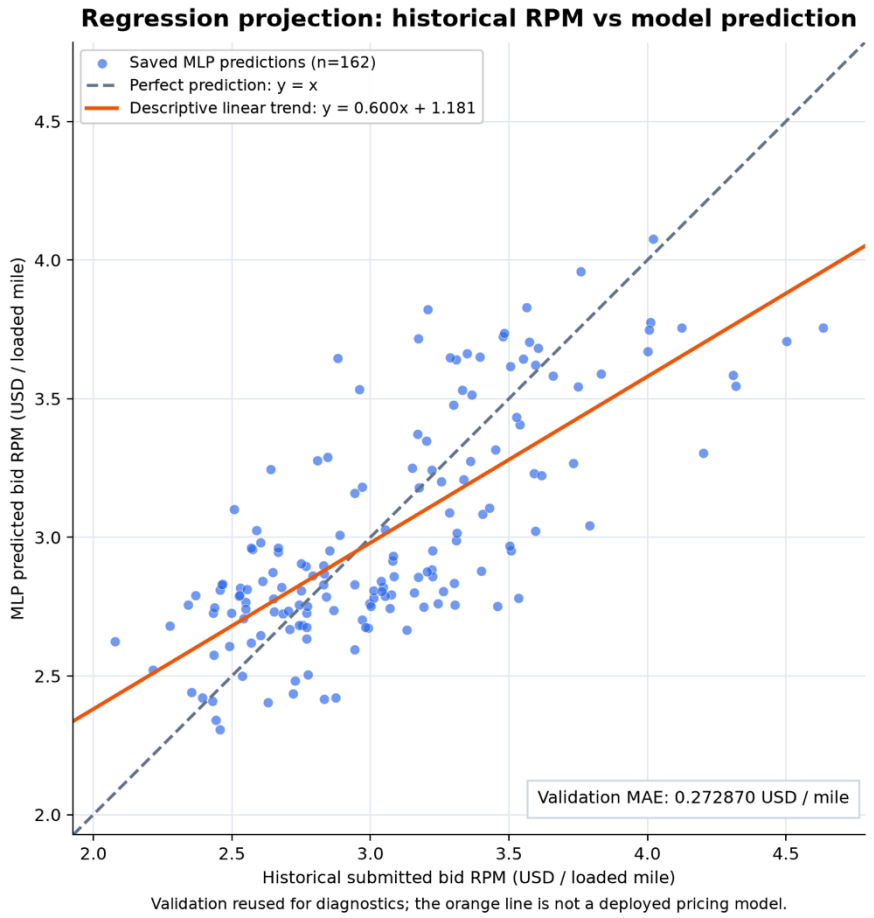
One-hot encoding turns route and state names into numbers the model can understand.





Predicting Number

The model learns from shipment details and historical submitted bids to estimate dollars per loaded mile (RPM). Multiplying that by the shipment's loaded miles gives the suggested total bid.

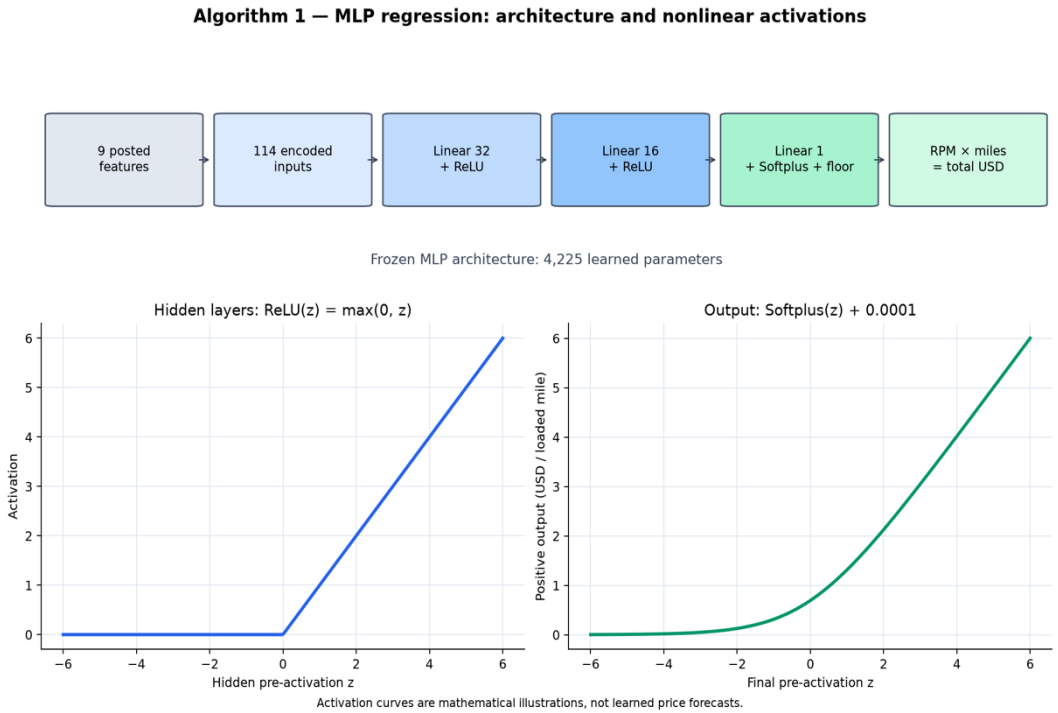


Ask the model to estimate a price using ReLU

Neural-network regression combines the shipment details to predict RPM.

Inside the network, ReLU helps it learn complex relationships.

For example, how distance and route together influence price. Softplus keeps the final predicted RPM positive.

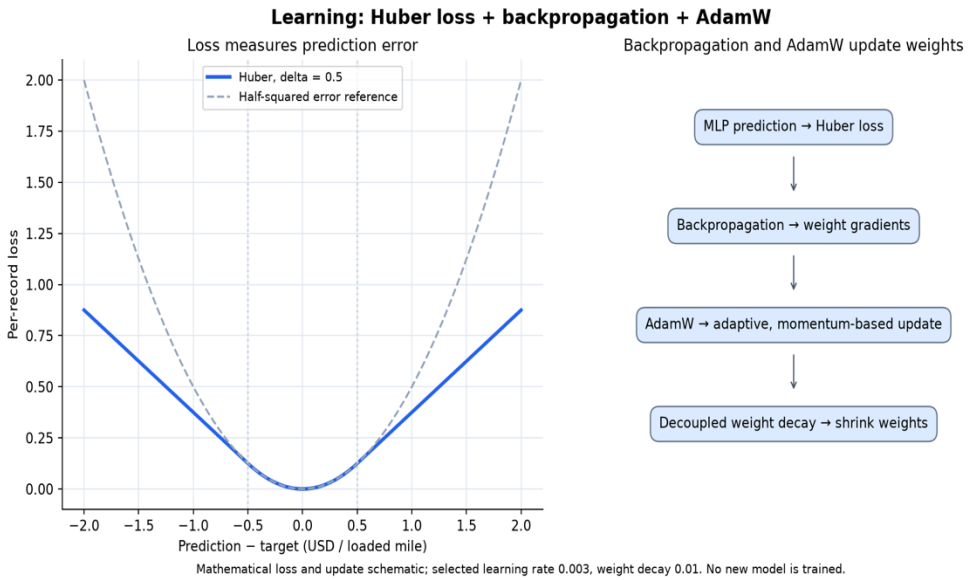


Learn from the mistake

The historical answer is calculated as:

Submitted bid ÷ loaded miles = historical RPM

Huber loss measures the difference between the prediction and that answer, while preventing a few extreme errors from dominating the learning.

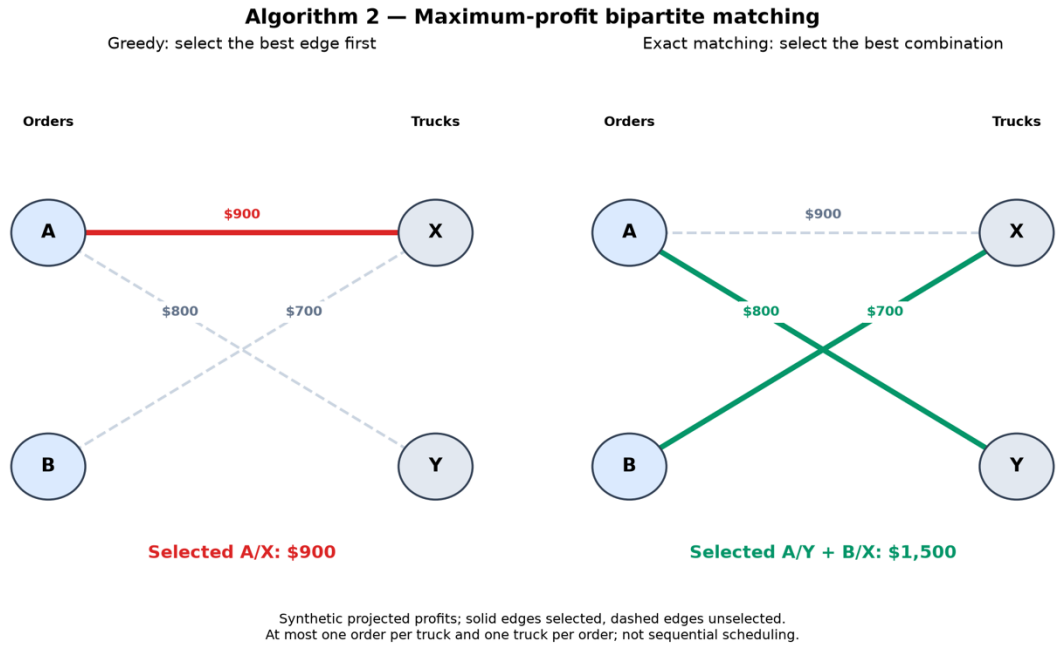


Repeat, check and save…

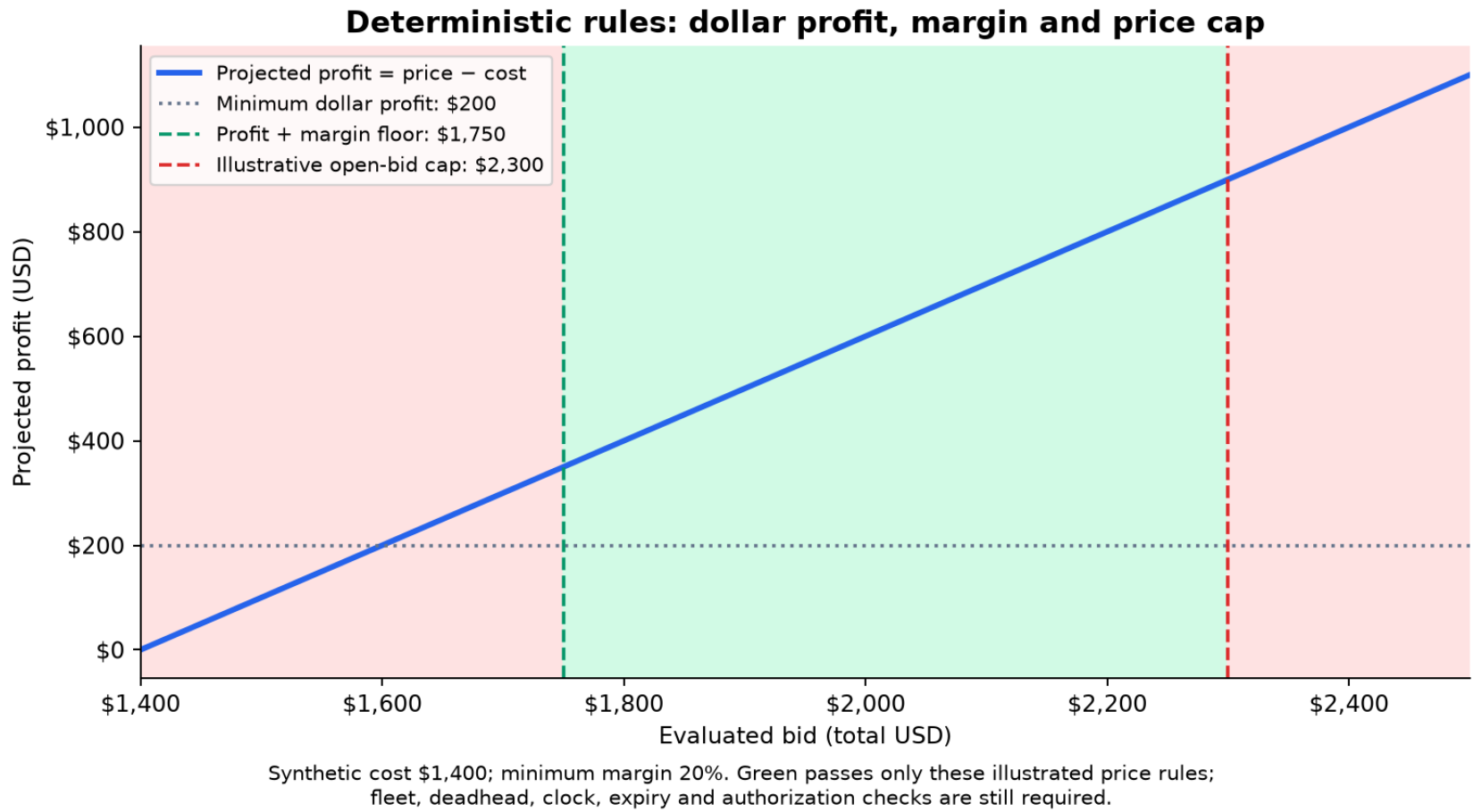
Live use of the model

Fleet matching

Inside the fleet matcher, successive shortest augmenting paths helps it choose the order truck combination with the highest total projected profit.



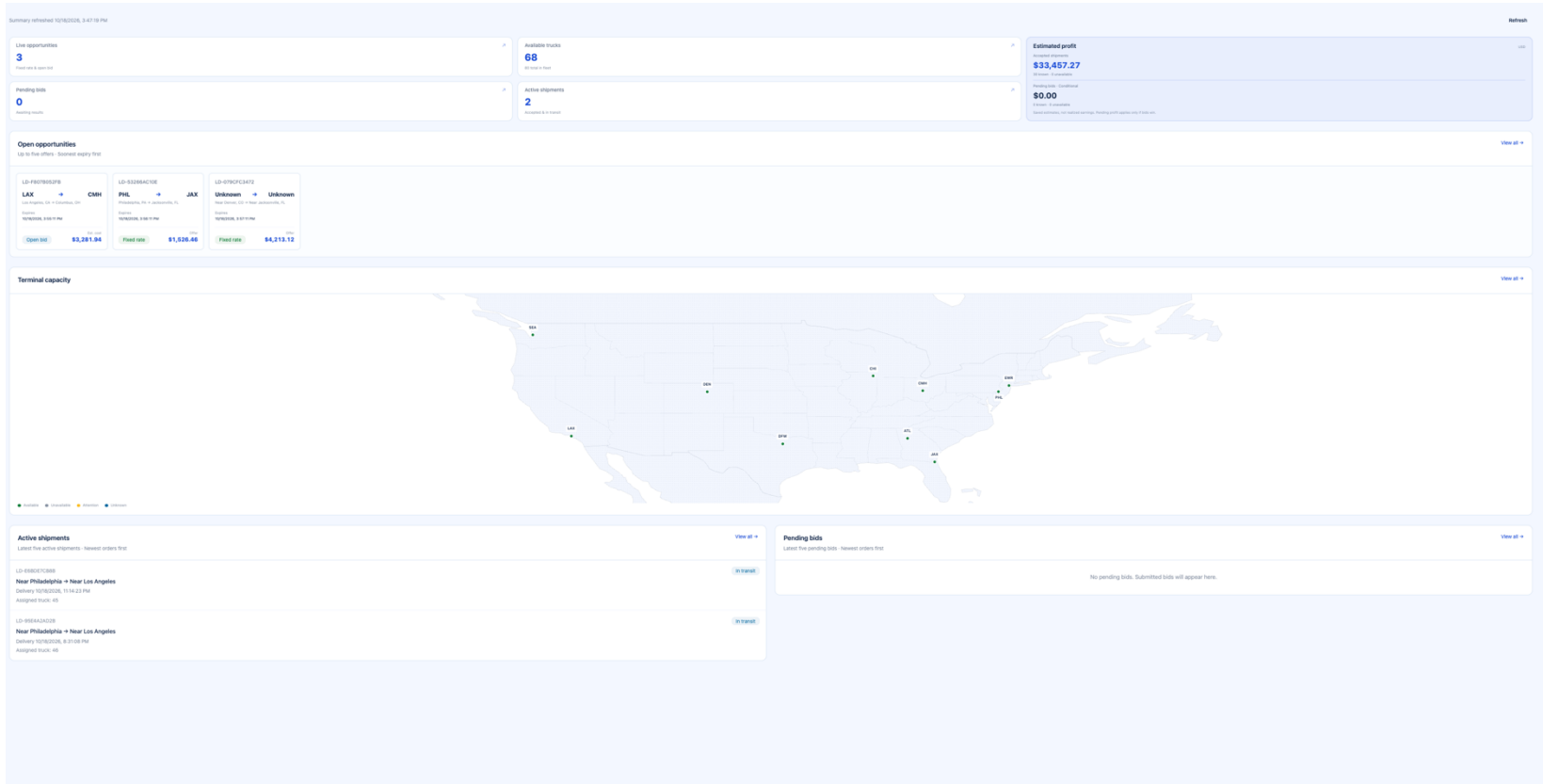
Calculate Bids



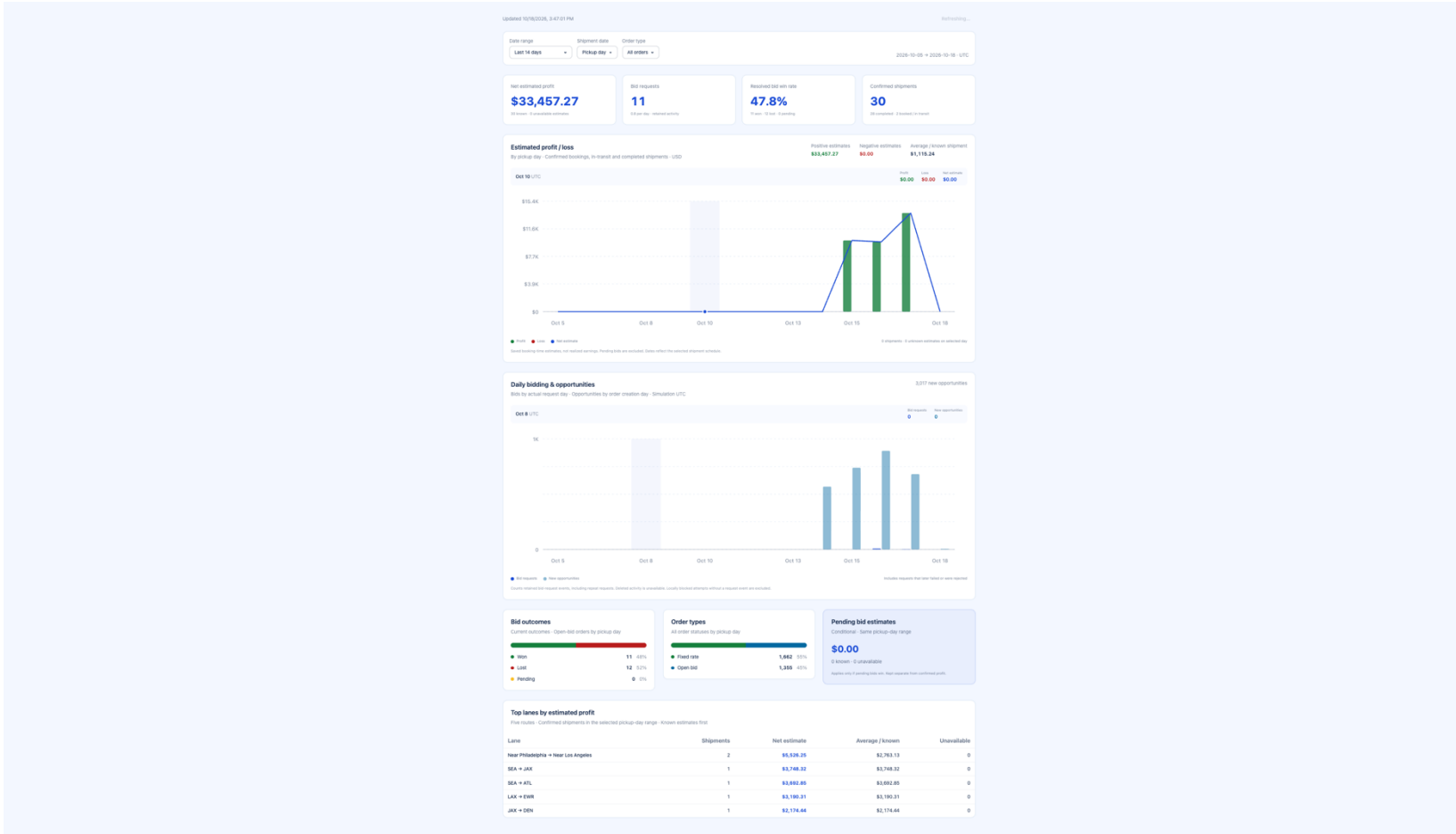
Synthetic cost $1,400; minimum margin 20%. Green passes only these illustrated price rules; fleet, deadhead, clock, expiry and authorization checks are still required.

Frontend Pages

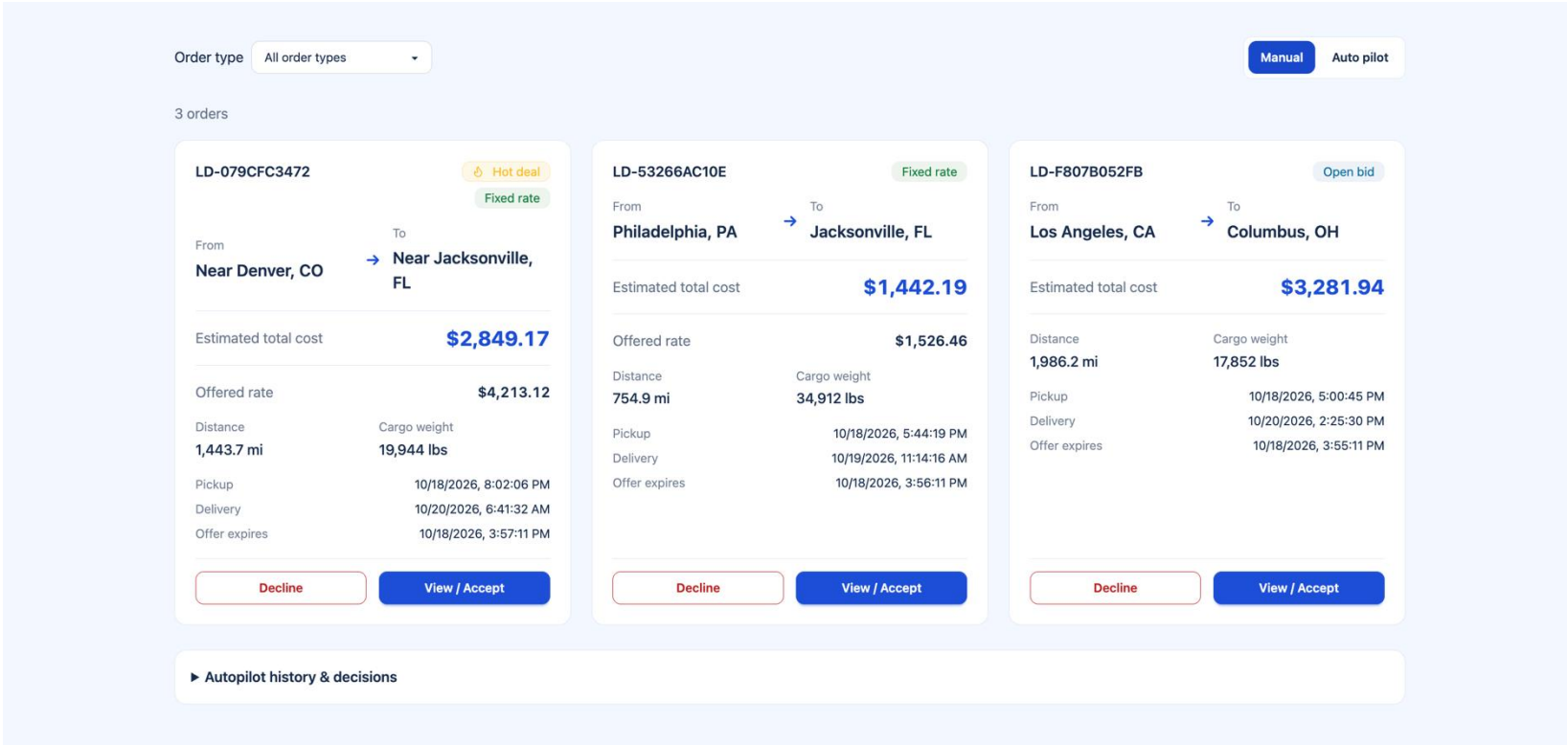
Dashboard


Summary refreshed 10/18/2026, 3:47:19 PM
Refresh
Live opportunities
3
Fixed rate & open bid
Available trucks
68
All time in fleet
Estimated profit
Accepted shipments
$33,457.27
30 known · 0 unavailable
Pending bids · Conditional
$0.00
0 known · 0 unavailable
Pending profit applies only if bids win.
Pending bids
0
Awaiting results
Active shipments
2
Accepted & in transit
Open opportunities
Up to five offers · Soonest expiry first
View all →
LD-F807B053FB
LAX → CMH
Los Angeles, CA → Columbus, OH
Expires
10/18/2026, 3:55:11 PM
Est. value
Open bid
$3,281.94
LD-53266AC10E
PHL → JAX
Philadelphia, PA → Jacksonville, FL
Expires
10/18/2026, 3:55:11 PM
Est. value
Fixed rate
$1,526.46
LD-079CFC3472
Unknown → Unknown
Near Denver, CO → Near Jacksonville, FL
Expires
10/18/2026, 3:57:11 PM
Est. value
Fixed rate
$4,213.12
Terminal capacity
View all →
SEA
DEN
CHI
CMH
EWR
PHL
LAX
DFW
ATL
JAX
Available
Unavailable
Attention
Unknown
Active shipments
Latest five active shipments · Newest orders first
View all →
LD-E68DE7C868
In transit
Near Philadelphia → Near Los Angeles
Delivery 10/18/2026, 11:14:23 PM
Assigned truck: 45
LD-9564A2A02B
In transit
Near Philadelphia → Near Los Angeles
Delivery 10/18/2026, 8:31:08 PM
Assigned truck: 46
Pending bids
Latest five pending bids · Newest orders first
View all →
No pending bids. Submitted bids will appear here.

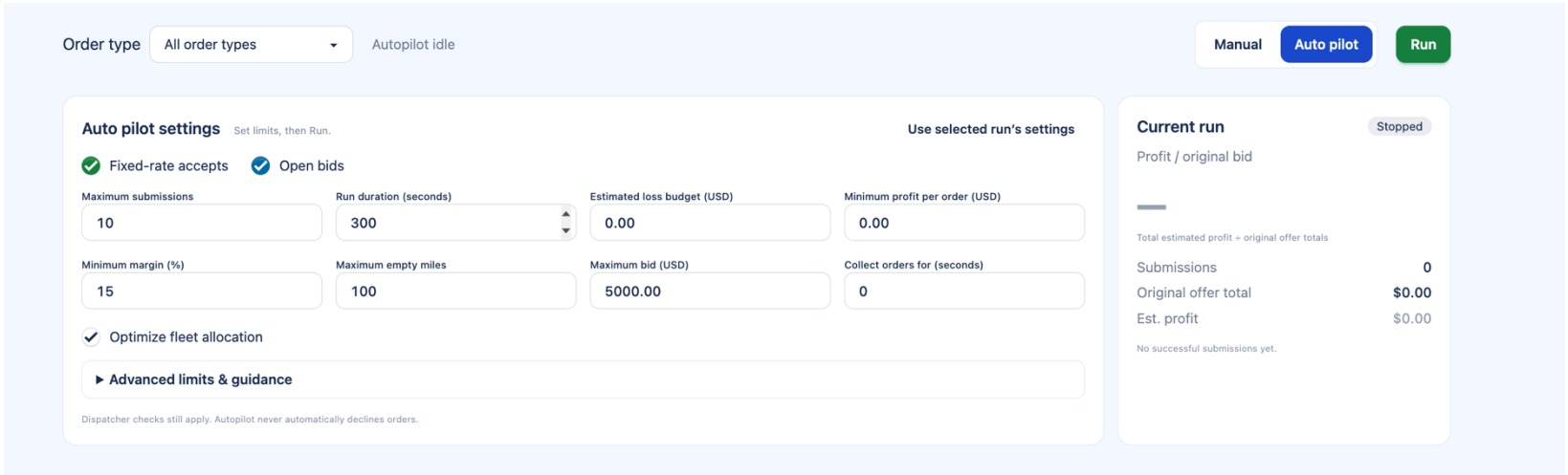
Analytics


Updated 10/18/2026, 3:47:01 PM
Refreshing...
Date range
Last 14 days
Shipment date
Pickup day
Order type
All orders
2026-10-05 → 2026-10-18 · UTC
Net estimated profit
$33,457.27
Bid requests
11
Resolved bid win rate
47.8%
Confirmed shipments
30
Estimated profit / loss
By pickup day · Confirmed bookings, in-transit and completed shipments · USD
Positive estimates
$33,457.27
Negative estimates
$0.00
Average / known shipment
$1,115.24
Oct 10 UTC
Profit
$0.00
Loss
$0.00
Net estimate
$0.00
$15.4K
$11.6K
$7.7K
$3.9K
$0
Oct 5
Oct 8
Oct 10
Oct 13
Oct 15
Oct 18
Profit
Loss
Net estimate
0 shipments · 0 unknown estimates on selected day
Saved booking-time estimates, not realized earnings. Pending bids are excluded. Dates reflect the selected shipment schedule.
Daily bidding & opportunities
Bids by actual request day · Opportunities by order creation day · Simulation UTC
3,017 new opportunities
Oct 8 UTC
Bid requests
0
New opportunities
0
1K
0
Oct 5
Oct 8
Oct 10
Oct 13
Oct 15
Oct 18
Bid requests
New opportunities
Counts retained bid request events, including repeat requests. Deleted activity is unavailable. Locally blocked attempts without a request event are excluded.
Includes requests that later failed or were rejected
Bid outcomes
Current outcomes · Open-bid orders by pickup day
Won
11 48%
Lost
12 52%
Pending
0 0%
Order types
All order statuses by pickup day
Fixed rate
1,662 55%
Open bid
1,355 45%
Pending bid estimates
Conditional · Same pickup-day range
$0.00
0 known · 0 unavailable
Applies only if pending bids win. Kept separate from confirmed profit.
Top lanes by estimated profit
Five routes · Confirmed shipments in the selected pickup-day range · Known estimates first
Lane
Shipments
Net estimate
Average / known
Unavailable
Near Philadelphia → Near Los Angeles
2
$5,526.25
$2,763.13
0
SEA → JAX
1
$3,748.32
$3,748.32
0
SEA → ATL
1
$3,692.85
$3,692.85
0
LAX → EWR
1
$3,190.31
$3,190.31
0
JAX → DEN
1
$2,174.44
$2,174.44
0

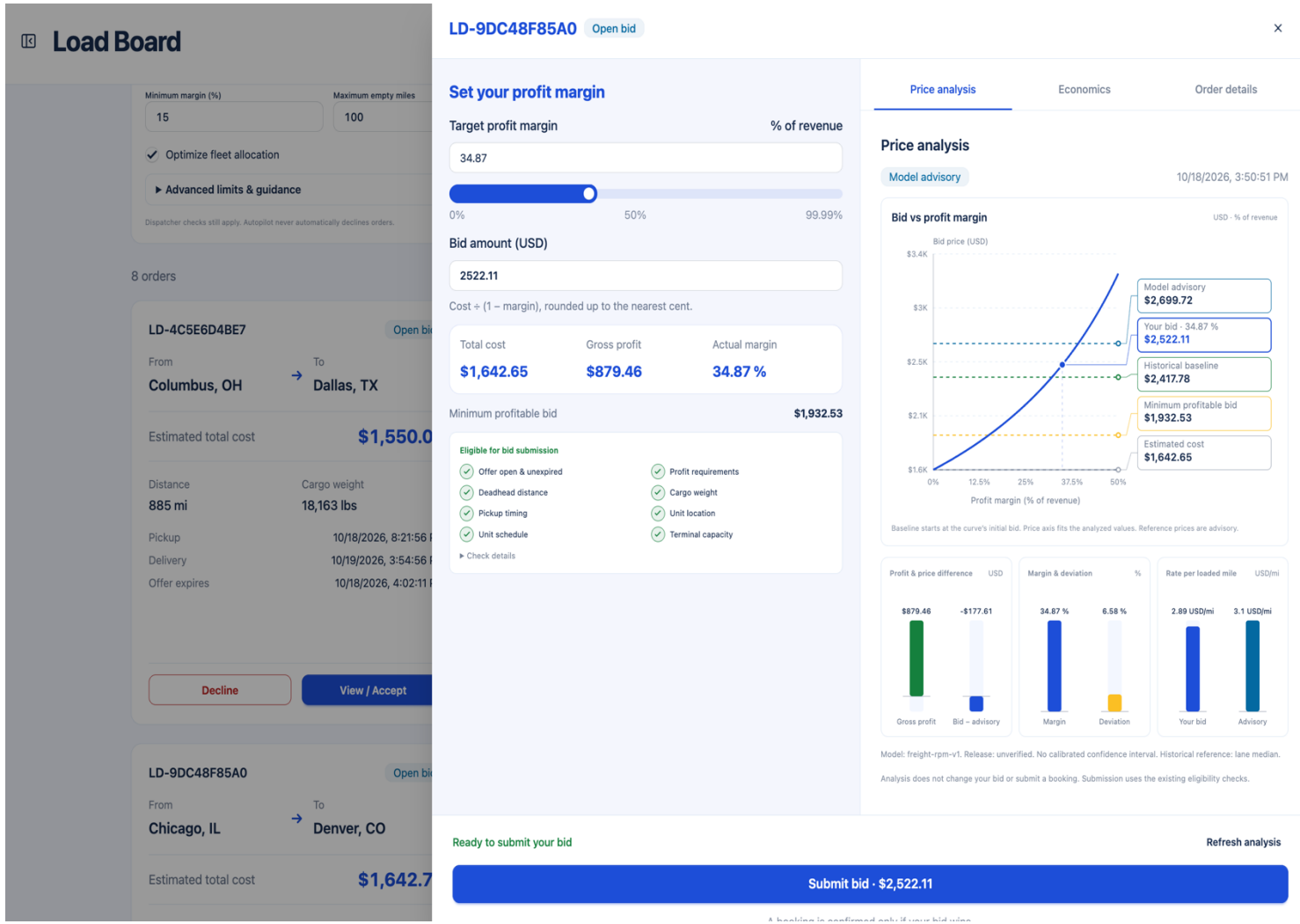
Load Board


Order type
All order types
Manual
Auto pilot
3 orders
LD-079CFC3472
Hot deal
Fixed rate
From
Near Denver, CO
To
Near Jacksonville, FL
Estimated total cost
$2,849.17
Offered rate
$4,213.12
Distance
1,443.7 mi
Cargo weight
19,944 lbs
Pickup
10/18/2026, 8:02:06 PM
Delivery
10/20/2026, 6:41:32 AM
Offer expires
10/18/2026, 3:57:11 PM
Decline
View / Accept
LD-53266AC10E
Fixed rate
From
Philadelphia, PA
To
Jacksonville, FL
Estimated total cost
$1,442.19
Offered rate
$1,526.46
Distance
754.9 mi
Cargo weight
34,912 lbs
Pickup
10/18/2026, 5:44:19 PM
Delivery
10/19/2026, 11:14:16 AM
Offer expires
10/18/2026, 3:56:11 PM
Decline
View / Accept
LD-F807B052FB
Open bid
From
Los Angeles, CA
To
Columbus, OH
Estimated total cost
$3,281.94
Distance
1,986.2 mi
Cargo weight
17,852 lbs
Pickup
10/18/2026, 5:00:45 PM
Delivery
10/20/2026, 2:25:30 PM
Offer expires
10/18/2026, 3:55:11 PM
Decline
View / Accept
▸ Autopilot history & decisions

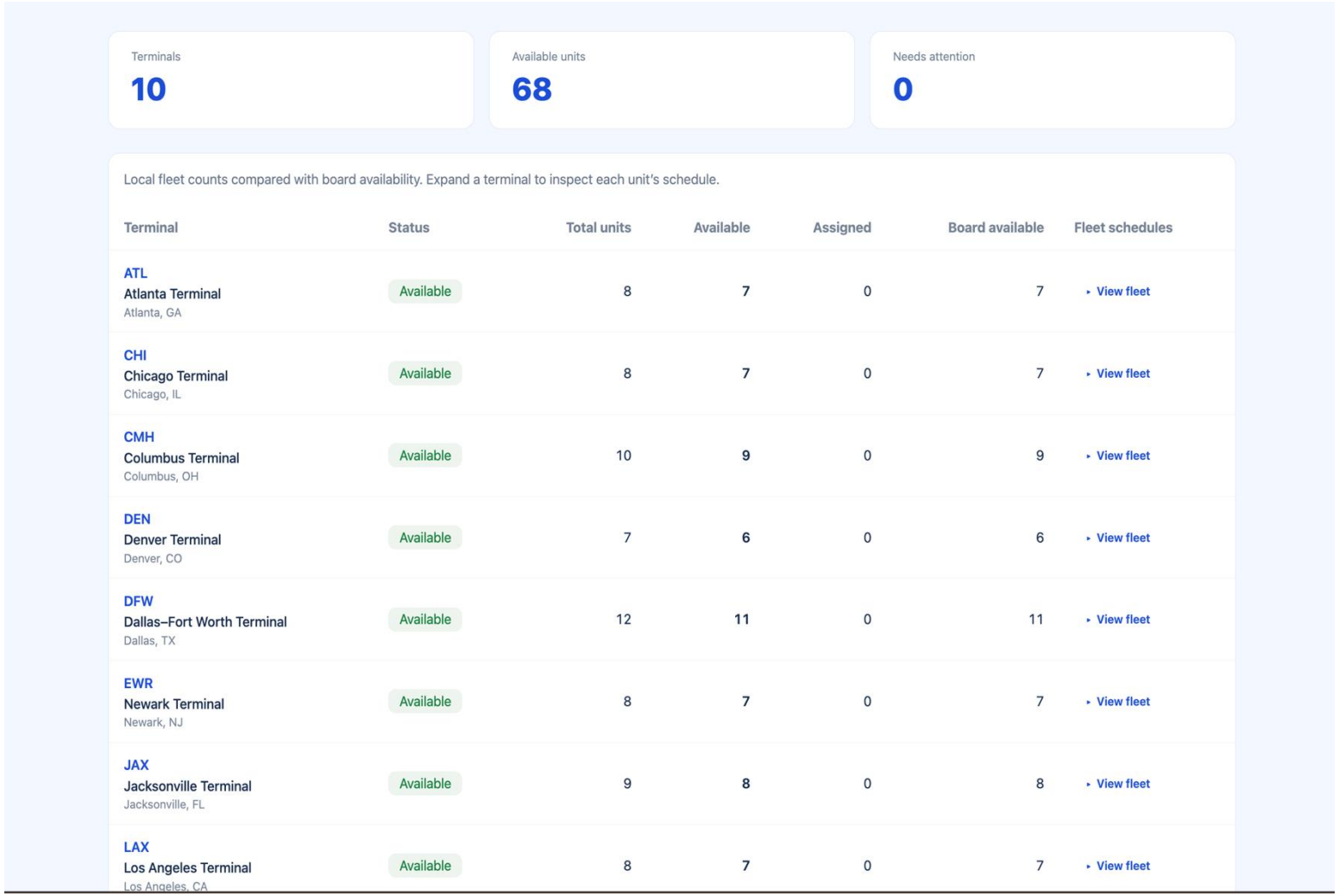
Auto-pilot limiter and settings



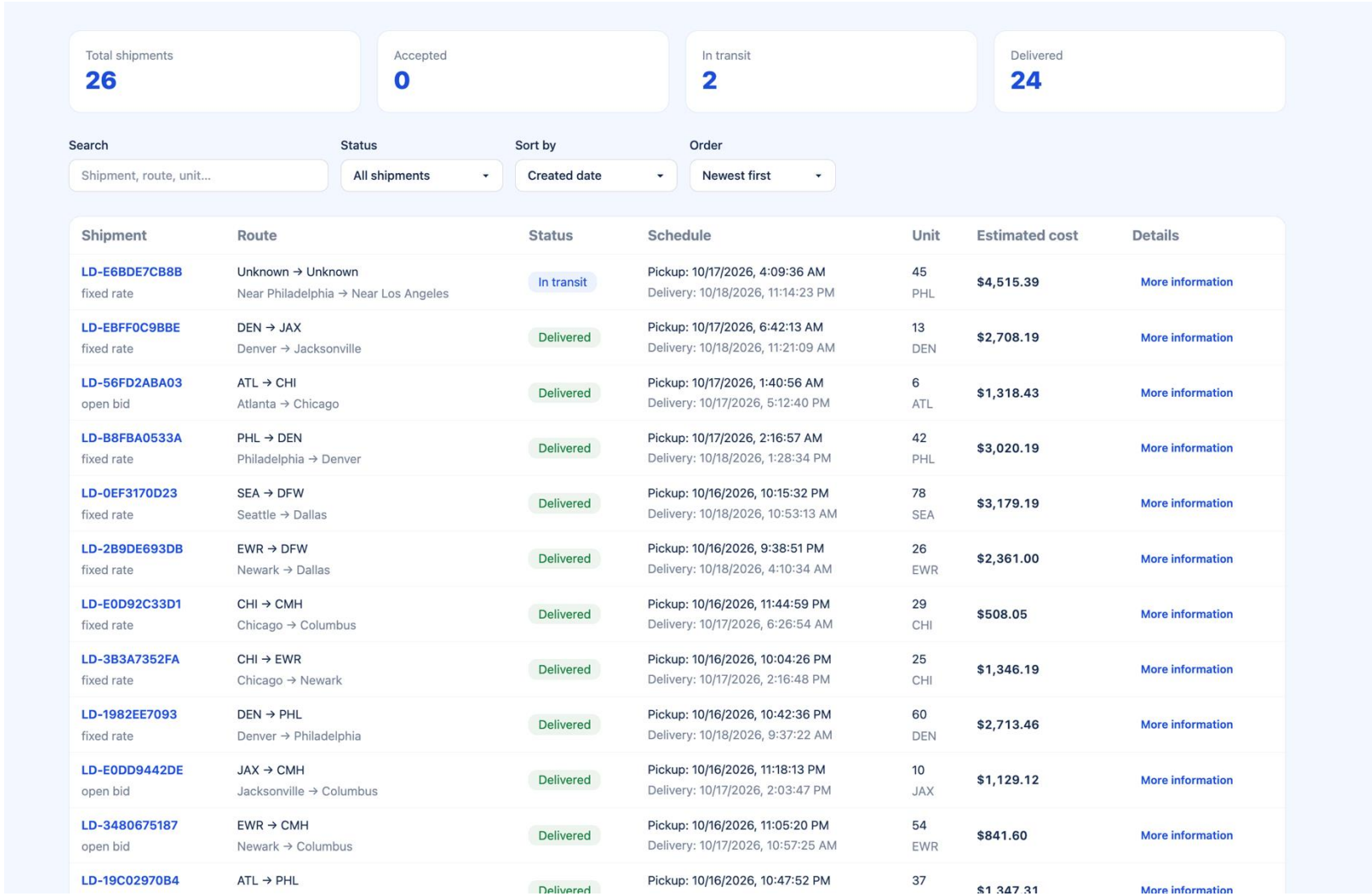
Price Analysis



Terminals


Terminals
10
Available units
68
Needs attention
0
Local fleet counts compared with board availability. Expand a terminal to inspect each unit's schedule.
Terminal
Status
Total units
Available
Assigned
Board available
Fleet schedules
ATL
Atlanta Terminal
Atlanta, GA
Available
8
7
0
7
View fleet
CHI
Chicago Terminal
Chicago, IL
Available
8
7
0
7
View fleet
CMH
Columbus Terminal
Columbus, OH
Available
10
9
0
9
View fleet
DEN
Denver Terminal
Denver, CO
Available
7
6
0
6
View fleet
DFW
Dallas–Fort Worth Terminal
Dallas, TX
Available
12
11
0
11
View fleet
EWR
Newark Terminal
Newark, NJ
Available
8
7
0
7
View fleet
JAX
Jacksonville Terminal
Jacksonville, FL
Available
9
8
0
8
View fleet
LAX
Los Angeles Terminal
Los Angeles, CA
Available
8
7
0
7
View fleet

Shipments


Total shipments
26
Accepted
0
In transit
2
Delivered
24
Search
Shipment, route, unit...
Status
All shipments
Sort by
Created date
Order
Newest first
Shipment
Route
Status
Schedule
Unit
Estimated cost
Details
LD-E6BDE7CB8B
fixed rate
Unknown → Unknown
Near Philadelphia → Near Los Angeles
In transit
Pickup: 10/17/2026, 4:09:36 AM
Delivery: 10/18/2026, 11:14:23 PM
45
PHL
$4,515.39
More information
LD-EBFF0C9BBE
fixed rate
DEN → JAX
Denver → Jacksonville
Delivered
Pickup: 10/17/2026, 6:42:13 AM
Delivery: 10/18/2026, 11:21:09 AM
13
DEN
$2,708.19
More information
LD-56FD2ABA03
open bid
ATL → CHI
Atlanta → Chicago
Delivered
Pickup: 10/17/2026, 1:40:56 AM
Delivery: 10/17/2026, 5:12:40 PM
6
ATL
$1,318.43
More information
LD-B8FBA0533A
fixed rate
PHL → DEN
Philadelphia → Denver
Delivered
Pickup: 10/17/2026, 2:16:57 AM
Delivery: 10/18/2026, 1:28:34 PM
42
PHL
$3,020.19
More information
LD-0EF3170D23
fixed rate
SEA → DFW
Seattle → Dallas
Delivered
Pickup: 10/16/2026, 10:15:32 PM
Delivery: 10/18/2026, 10:53:13 AM
78
SEA
$3,179.19
More information
LD-2B9DE693DB
fixed rate
EWR → DFW
Newark → Dallas
Delivered
Pickup: 10/16/2026, 9:38:51 PM
Delivery: 10/18/2026, 4:10:34 AM
26
EWR
$2,361.00
More information
LD-E0D92C33D1
fixed rate
CHI → CMH
Chicago → Columbus
Delivered
Pickup: 10/16/2026, 11:44:59 PM
Delivery: 10/17/2026, 6:26:54 AM
29
CHI
$508.05
More information
LD-3B3A7352FA
fixed rate
CHI → EWR
Chicago → Newark
Delivered
Pickup: 10/16/2026, 10:04:26 PM
Delivery: 10/17/2026, 2:16:48 PM
25
CHI
$1,346.19
More information
LD-1982EE7093
fixed rate
DEN → PHL
Denver → Philadelphia
Delivered
Pickup: 10/16/2026, 10:42:36 PM
Delivery: 10/18/2026, 9:37:22 AM
60
DEN
$2,713.46
More information
LD-E0DD9442DE
open bid
JAX → CMH
Jacksonville → Columbus
Delivered
Pickup: 10/16/2026, 11:18:13 PM
Delivery: 10/17/2026, 2:03:47 PM
10
JAX
$1,129.12
More information
LD-3480675187
open bid
EWR → CMH
Newark → Columbus
Delivered
Pickup: 10/16/2026, 11:05:20 PM
Delivery: 10/17/2026, 10:57:25 AM
54
EWR
$841.60
More information
LD-19C02970B4
ATL → PHL
Delivered
Pickup: 10/16/2026, 10:47:52 PM
37
$1,347.31
More information